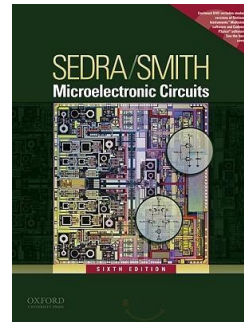


Partie 1 : Dispositifs et Circuits de base

- 1 : Signaux et Amplificateurs
 - 2 : Amplificateurs opérationnels
- (Rappels :
Un bon ampli = ?)



Chapters 1 & 2



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ELEC1530 – Partie 1 : Signaux et Amplificateurs - 1

1

Introduction : Inside a Smartphone



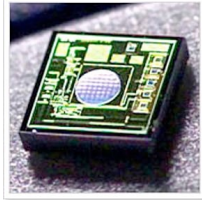
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ELEC1530 – Partie 1 : Signaux et Amplificateurs - 2

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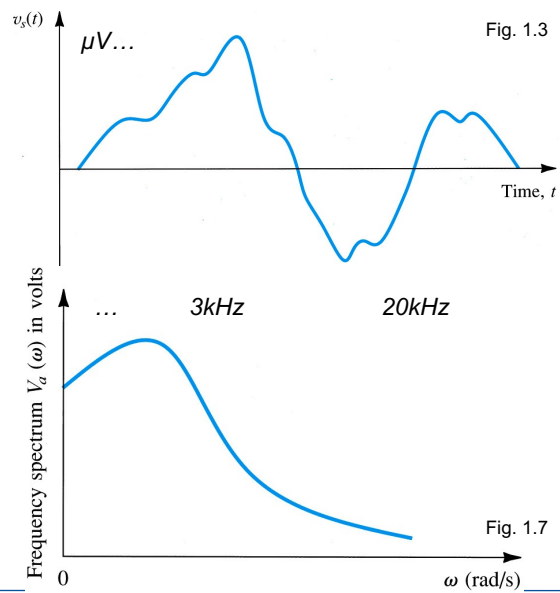
1.1-2 : Signaux

Microphone MEMS



Signal analogique

- amplitude (V)
- temps (s)
- contenu
fréquentiel



1.1-2 : Sources (de signaux)

Microphone MEMS, antenne RF, capteur d'orientation...

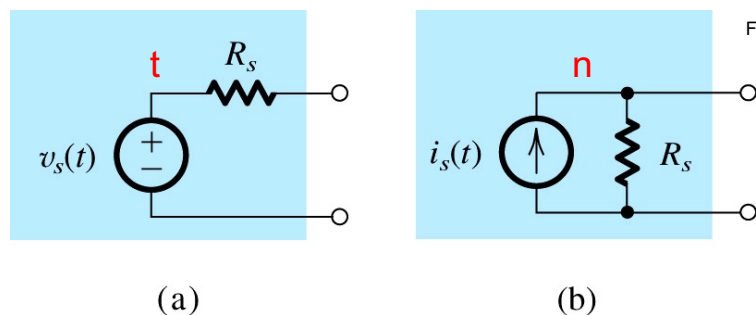
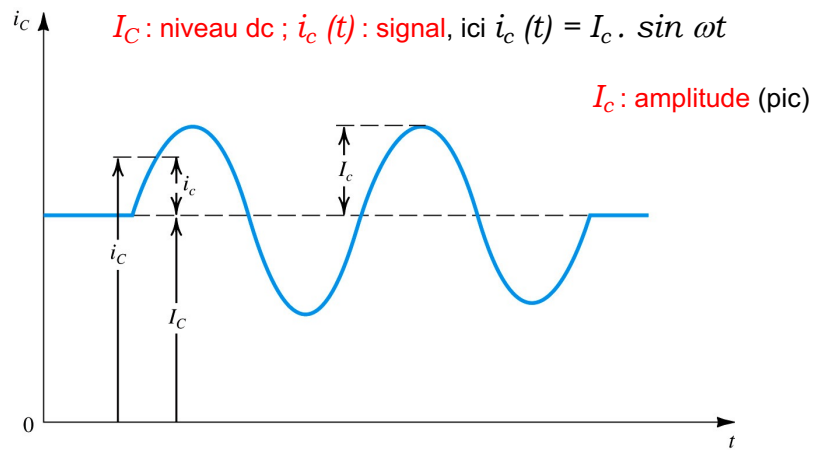


Fig. 1.1

Equivalents de
(a) Thévenin et (b) Norton

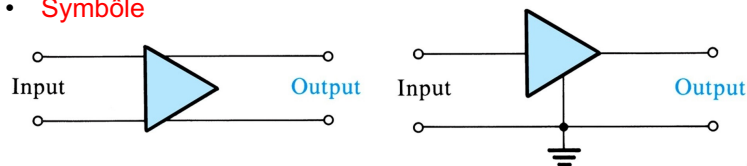
Conventions pour courants et tensions (1.4.8)

$i_c(t) = I_C + i_c(t)$: courant total instantané (**dc + ac**)

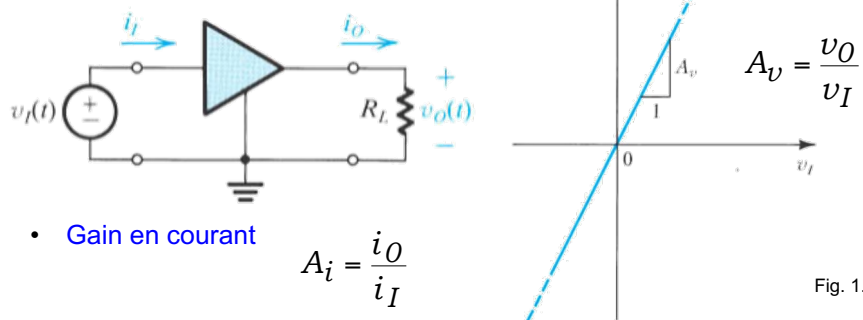


1.4 : Amplificateurs

- **Symbole**



- **Gain en tension (A_v)**



- **Gain en courant**

$$A_i = \frac{i_O}{i_I}$$

- **Gain en puissance** $A_p = A_v \cdot A_i = \frac{v_o \cdot i_o}{v_i \cdot i_i} = \frac{P_L}{P_I}$ $L = \text{'load'}$
(charge)
- **Gain en V en dB** $20 \cdot \log |A_v|$ [dB]
(rem. : $10 \cdot \log |A_p|$)
- **Rendement ('efficiency')** $\eta = \frac{P_L}{P_{dc}} \times 100$ [%]

$$P_{dc} = V_{CC} \cdot I_{CC} + V_{EE} \cdot I_{EE}$$

$$P_{dc} + P_I = P_L + P_{dissipée}$$

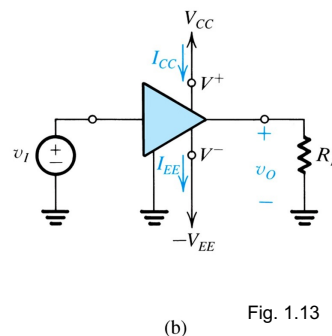
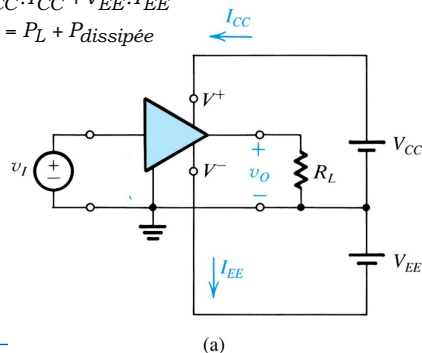


Fig. 1.13

Plan du 1^{er} cours

- 1 : Signaux : Sources, Analogiques / Digitaux, Notations
- 2 : Amplificateurs opérationnels (AO)
(et notions d'amplificateurs du chapitre 1)
Cas d'étude : AO de type $\mu A741$ – Simulations SPICE
2.1 : AO « idéal » et imperfections
 - Signaux d'entrée différentiels et de mode commun
 - DC : gain, limites, offset, courants d'entrée (2.6)
 - AC : réponse en fréquence (1.6, 2.7)

B.9. Cas d'étude : Amplificateur $\mu A741$

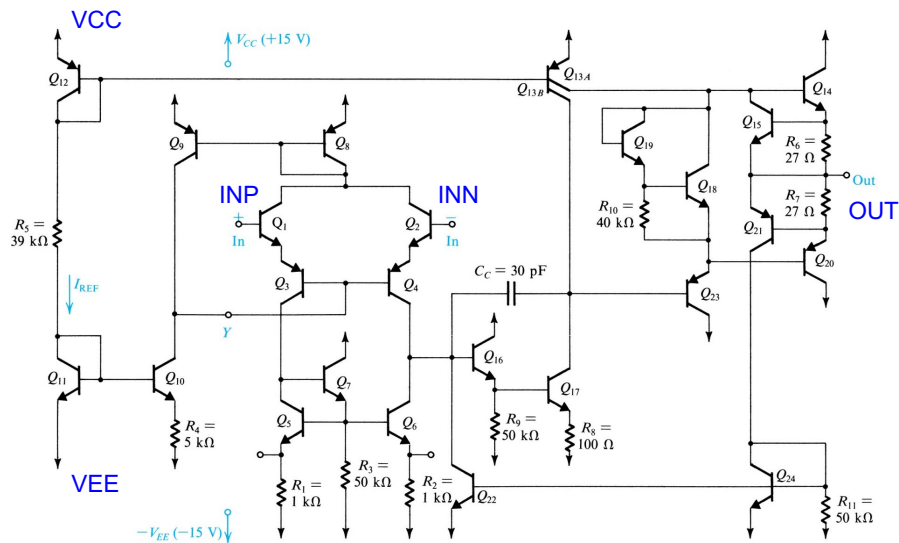


Fig. 12.13

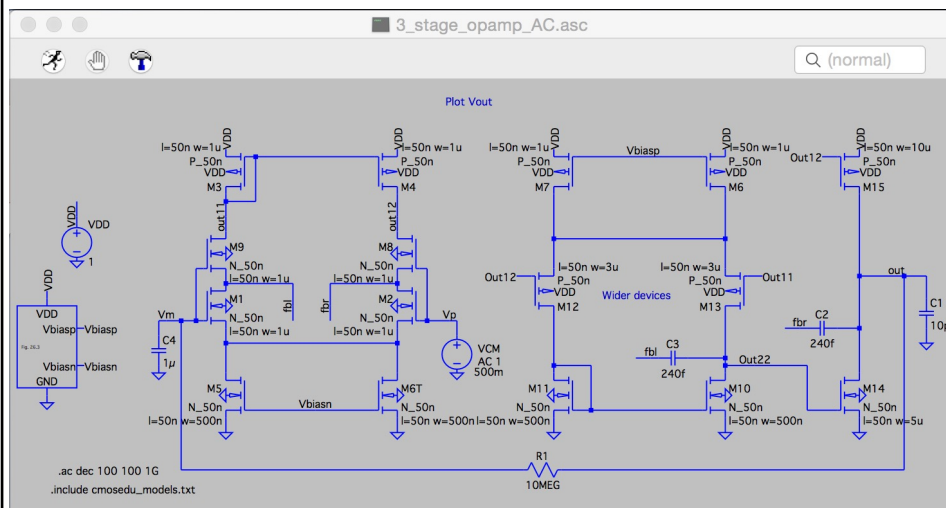


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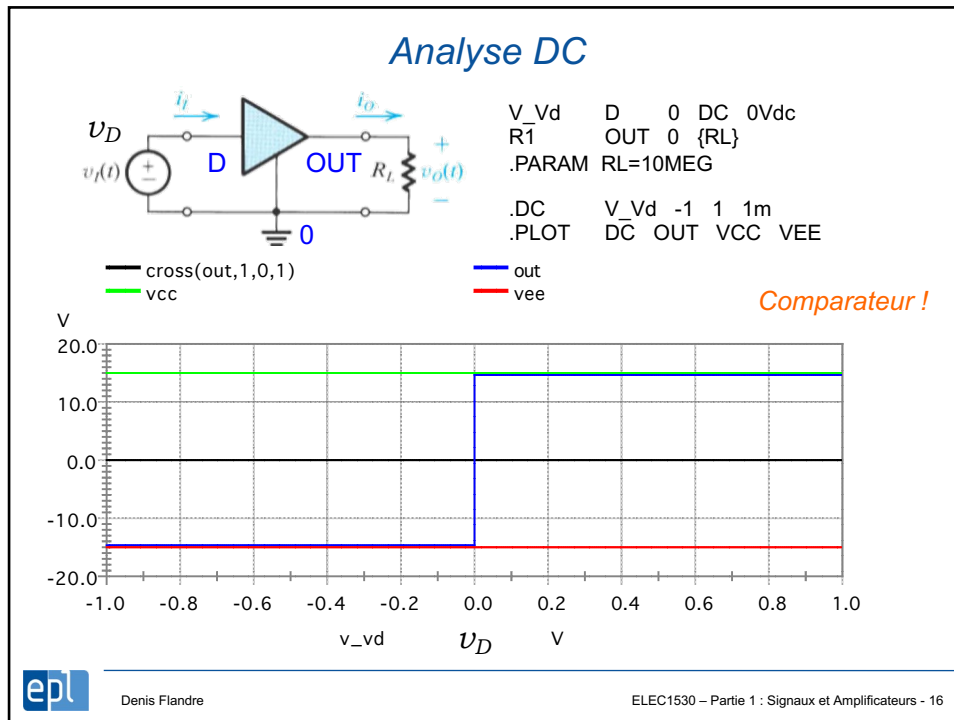
Simulation de circuits avec SPICE



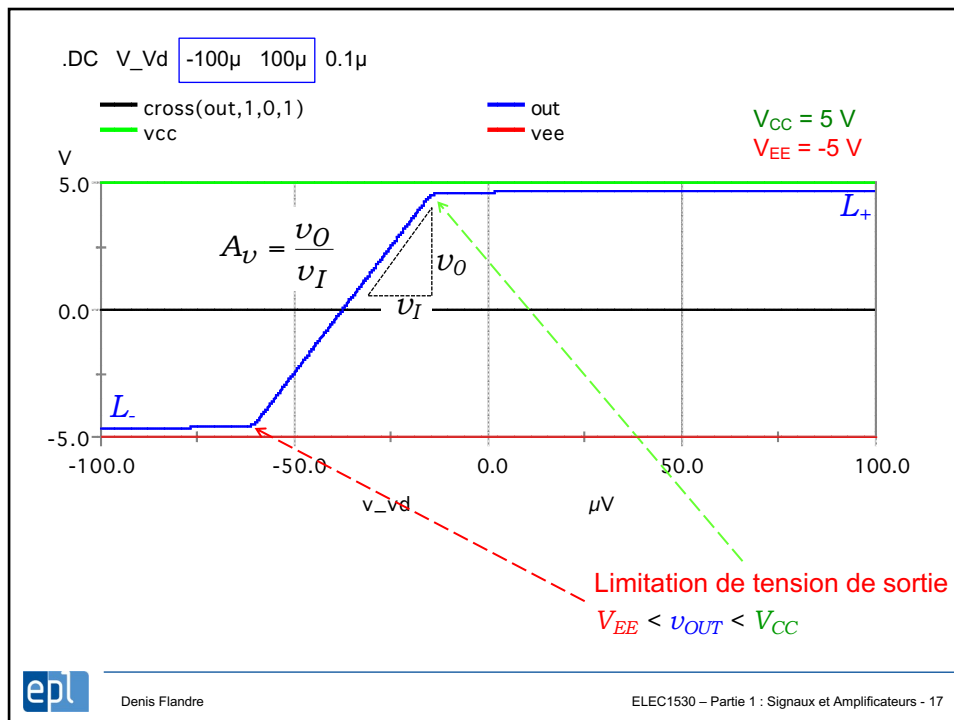
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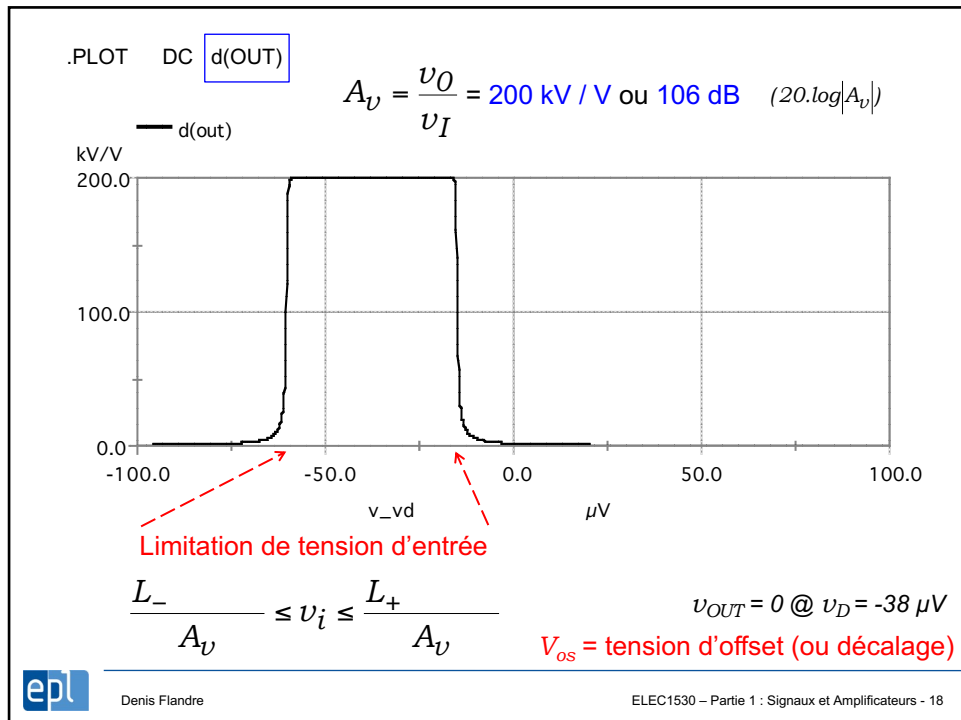
12



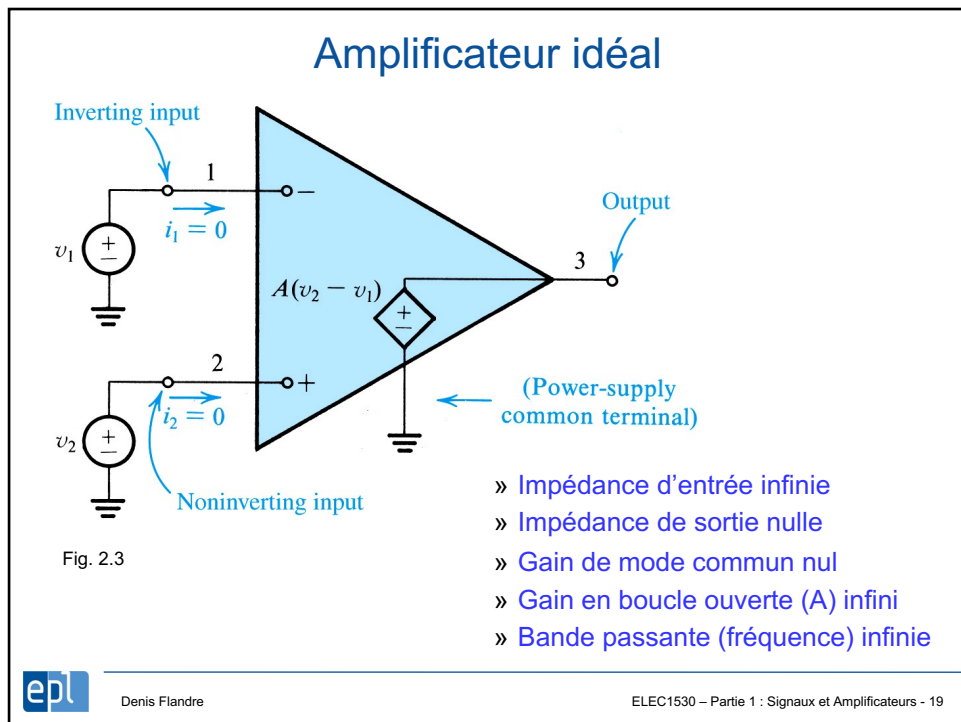
16



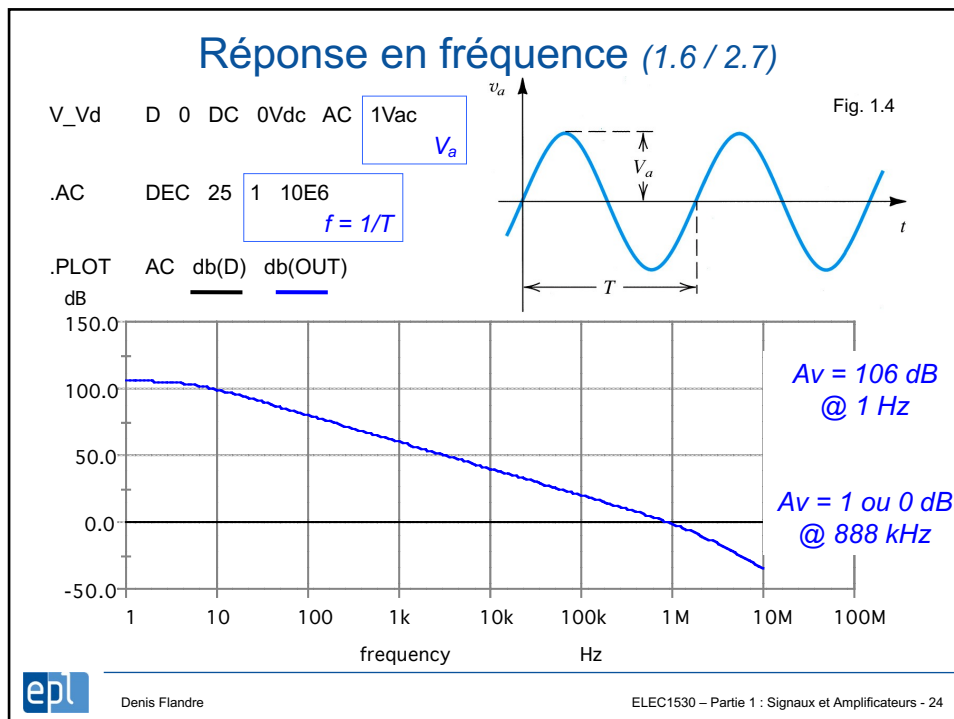
17



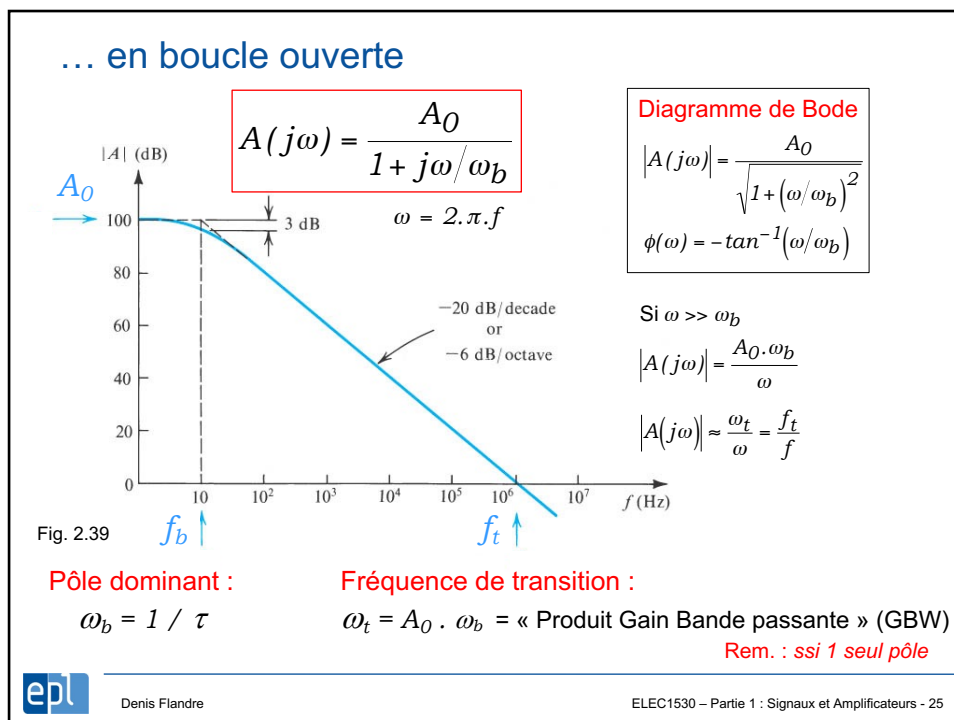
18



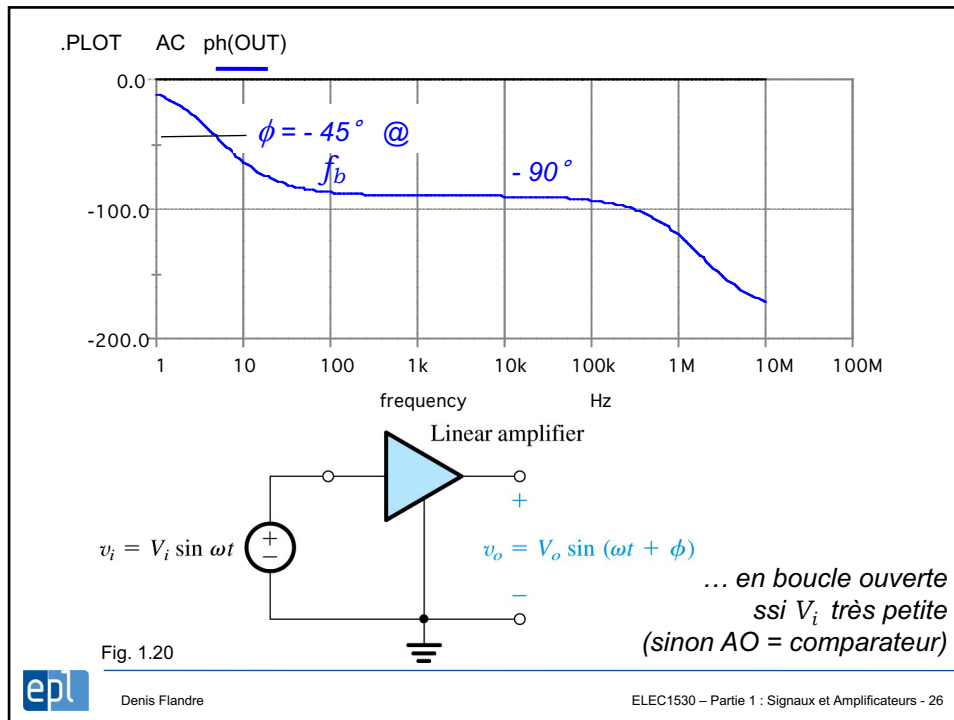
19



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... en boucle fermée

Configurations : 'inverseuse' (inverting)
et 'non-inverseuse' (non-inverting)

Ex. : Amplificateur de gain unitaire ou Suiveur de tension ou 'Buffer'

(a)

(b)

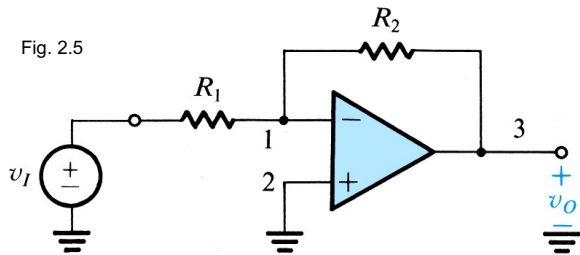
Fig. 2.14

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2.2. Inverting configuration

Fig. 2.5



$$G \equiv \frac{v_O}{v_I}$$

$$v_2 - v_1 = v_O / A \approx 0$$

("masse virtuelle")

$$i_I \approx$$

$$v_O \approx$$

$$G \approx$$

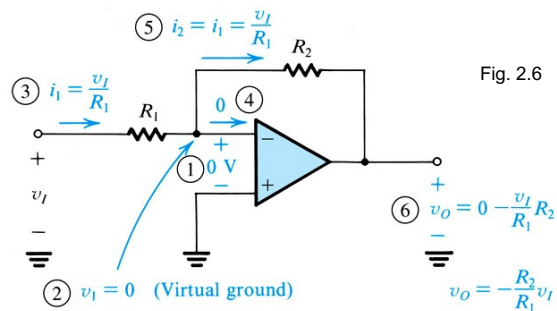


Fig. 2.6



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Effet de la réponse en fréquence de l'AO (2.6.2)

$$G = \frac{-R_2 / R_1}{1 + (1 + R_2 / R_1) / A(s)}$$

$$A(s) = \frac{A_0}{1 + s / \omega_b}$$

Si $A_0 \gg 1 + R_2 / R_1$

$$\frac{V_O(s)}{V_I(s)} \approx \frac{-R_2 / R_1}{1 + \frac{s}{\omega_t (1 + R_2 / R_1)}}$$

pôle : $\omega_{3dB} = \omega_t / (1 + R_2 / R_1)$

Exemple : $R_2 / R_1 = 100$

$\Rightarrow G_0 \approx 40 \text{ dB}$
 $f_{3dB} \approx f_t / 100$

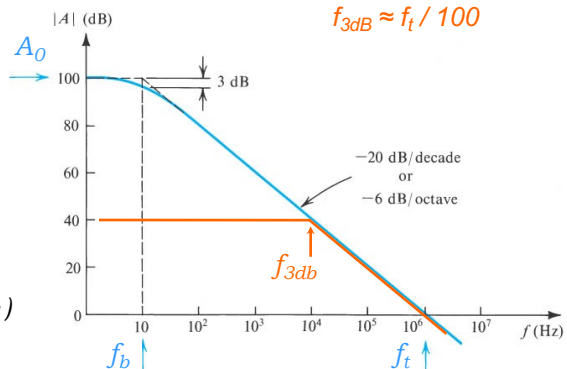


Fig. 2.39

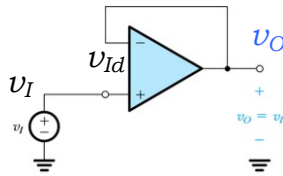


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2.3 : Non-inverting configuration

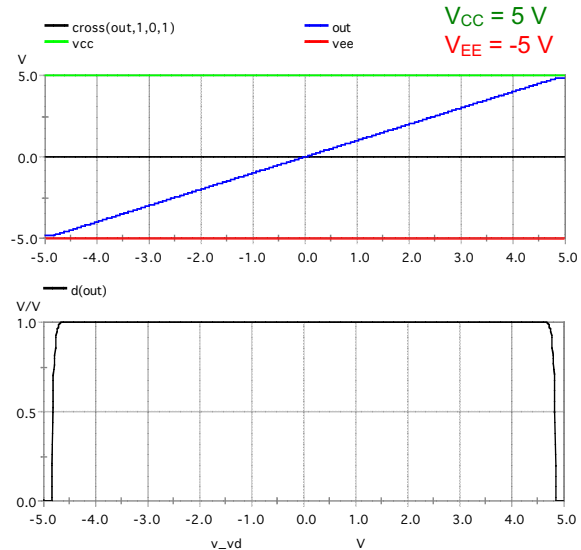


Analyse DC

$$v_O = \dots =$$

$$G = v_O / v_I =$$

- Plage d'entrée $\leftrightarrow$
- Gain ≈ 1



Analyse AC

$$v_O = A \cdot v_{Id} = A \cdot (v_I - v_O)$$

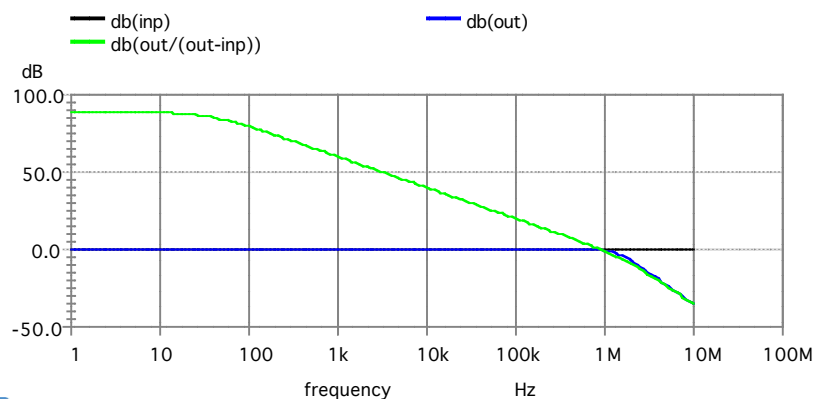
$$G = v_O / v_I = 1 / \left(1 + \frac{1}{A} \right)$$

$$\rightarrow A = v_O / v_{Id}$$

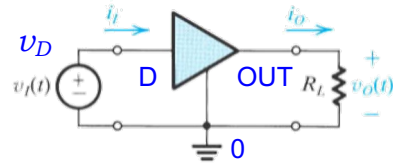
$$A < 106 \text{ dB ! ???}$$

$$A(s) = \frac{A_0}{1 + s/\omega_b} \quad (s = j\omega)$$

$$\Rightarrow G(s) \approx 1 / \left(1 + s/\omega_t \right) \quad \text{si } A_0 \uparrow \uparrow \quad \text{Bande passante } \leftrightarrow \text{ et } G \approx 1 \text{ jusque } f_t$$



Observation 1 (Influence de R_L en boucle ouverte)



R_L	A_v (V/V)	A_v (dB)
100 MOhm	191 600	105,6
100 Ohm	77 990	97,8

Observation 2 (Courant d'entrée en ac)

AC : amplitude $I_i \approx 1 \mu A$ @ $V_d = 1 V$

+ Résistances d'entrée et de sortie

$$G_v = A_{vo} \frac{R_i}{R_i + R_s} \frac{R_L}{R_L + R_o}$$

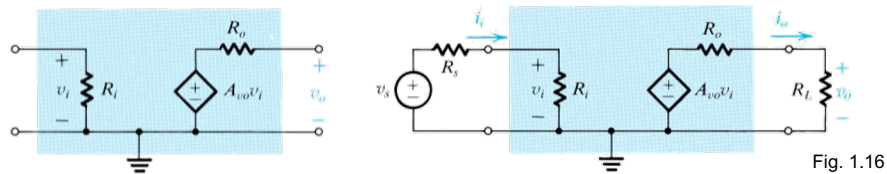


Fig. 1.16



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Modèles de circuits pour amplificateurs (1.5)

TABLE 1.1 The Four Amplifier Types

Type	Circuit Model	Gain Parameter	Ideal Characteristics
Voltage Amplifier		Open-Circuit Voltage Gain $A_{vo} \equiv \frac{v_o}{v_i} \Big _{i_o=0}$ (V/V)	$R_i = \infty$ $R_o = 0$
Current Amplifier		Short-Circuit Current Gain $A_{is} \equiv \frac{i_o}{i_i} \Big _{v_o=0}$ (A/A)	$R_i = 0$ $R_o = \infty$
Transconductance Amplifier		Short-Circuit Transconductance $G_m \equiv \frac{i_o}{v_i} \Big _{v_o=0}$ (A/V)	$R_i = \infty$ $R_o = \infty$
Transresistance Amplifier		Open-Circuit Transresistance $R_m \equiv \frac{v_o}{i_i} \Big _{v_o=0}$ (V/A)	$R_i = 0$ $R_o = 0$



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Amplificateurs avec étages en cascade

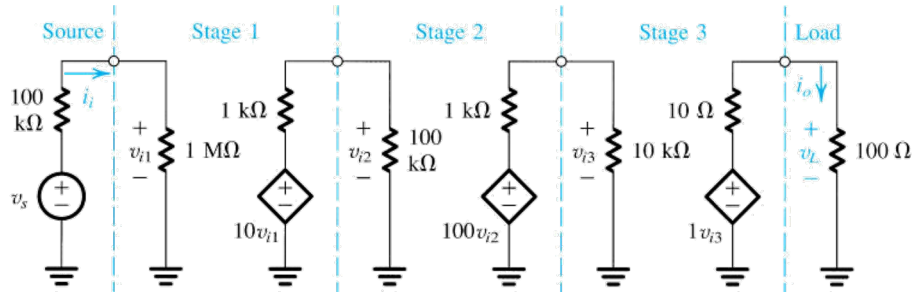


Fig. 1.17

$$R_s \ll R_{i,\text{étage } 1}$$

$$R_{o,\text{étage } i-1} \ll R_{i,\text{étage } i}$$

$$G_v = A_{vo} \frac{R_i}{R_i + R_s} \frac{R_L}{R_L + R_o}$$

$$R_{o,\text{étage } 3} \ll R_L$$



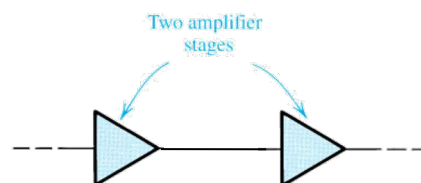
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• Etages successifs

» Directly coupled (dc)



$$\text{ssi dc : } V_{OUT,\text{étage } i-1} = V_{IN,\text{étage } i}$$

» Capacitively coupled (ac)

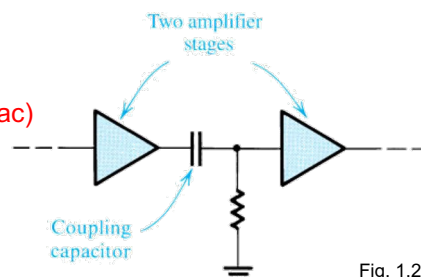


Fig. 1.27



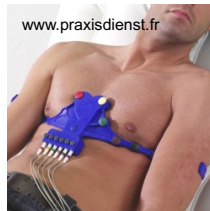
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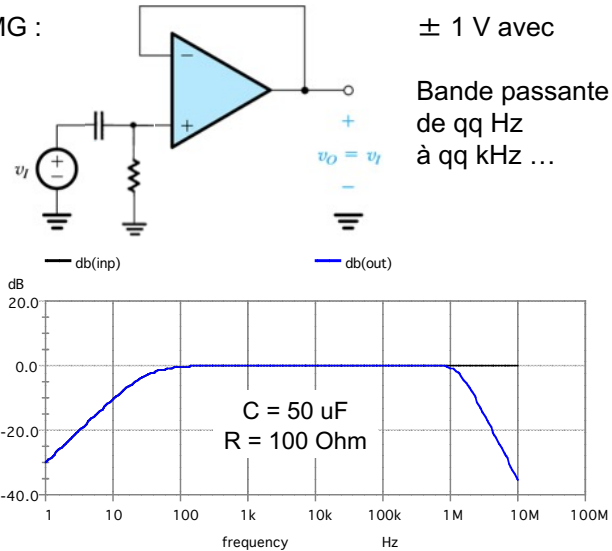
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Exemple : électrode médicale

Signaux ECG, EEG, EMG :
 $\mu\text{V} \dots \text{mV}$ ac, $\pm 15\text{V}$ dc
 Pas de contact DC
 avec patient



Medtech = secteur stratégique
 de spécialisation intelligente
 en Wallonie(ex. :
 Synergia Medical (LLN),
 Nomics (Liège)...)



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Single-Time-Constant Networks (STC)

Fig. 1.22

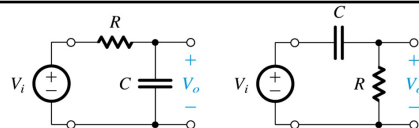


TABLE 1.2 Frequency Response of STC Networks

Filtres

	Low-Pass (LP)	High-Pass (HP)
Transfer Function $T(s)$	$\frac{K}{1 + (s/\omega_0)}$	$\frac{Ks}{s + \omega_0}$
Transfer Function (for physical frequencies) $T(j\omega)$	$\frac{K}{1 + j(\omega/\omega_0)}$	$\frac{K}{1 - j(\omega_0/\omega)}$
Magnitude Response $ T(j\omega) $	$\frac{ K }{\sqrt{1 + (\omega/\omega_0)^2}}$	$\frac{ K }{\sqrt{1 + (\omega_0/\omega)^2}}$
Phase Response $\angle T(j\omega)$	$-\tan^{-1}(\omega/\omega_0)$	$\tan^{-1}(\omega_0/\omega)$
Transmission at $\omega = 0$ (dc)	K	0
Transmission at $\omega = \infty$	0	K
3-dB Frequency	$\omega_0 = 1/\tau$; $\tau \equiv$ time constant $\tau = CR$ or L/R	
Bode Plots	in Fig. 1.23	in Fig. 1.24



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